

Predicting the Future Performance of the Planned Seismic Network in Chinese mainland

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Using the Bayesian Magnitude of Completeness (BMC) method, we generated the spatial distribution of M_c predicted for the new network, based on the prior model calibrated on the current earthquake catalog from January 1, 2009, to June 26, 2022, and network configuration. We further quantify the potential improvement of seismicity in a more complete catalog observed with the change in the network and investigate the significance for seismic-based earthquake prediction research based on the optimal use of data. This file serves as a valuable supplement to the text, providing a comprehensive collection of informative visuals and data analysis. It includes plots that showcase the distribution of earthquakes in Chinese mainland. Moreover, detailed maps are provided to depict the inter-station distances for the existing and planned networks, along with the standard deviation of the observed M_c values.

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Figure S1 to The epicenters and faults in China and its adjacent regions.

Figure S2 to Maps showing the inter-station distances for the existing and planned broadband seismic stations in China.

Figure S3 to Map of the standard deviation of the observed M_c .

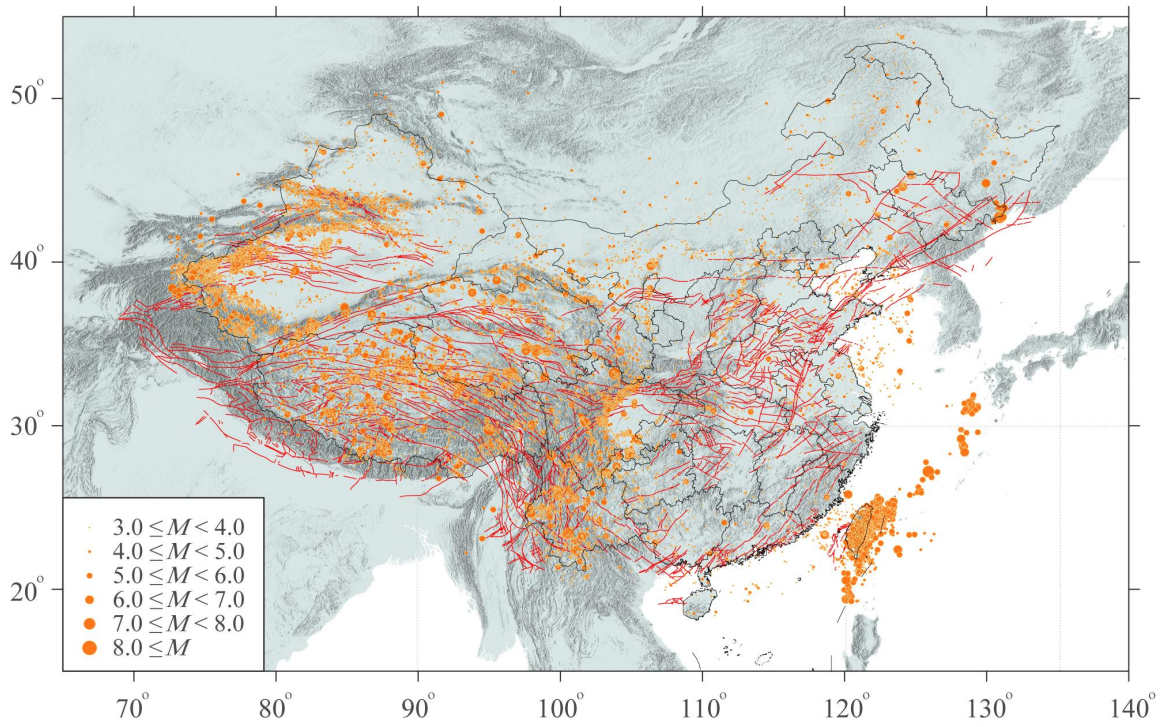


Figure S1. Distribution of 23,693 $M_L \geq 3.0$ earthquakes from 01/01/2009 to 26/06/2022 in China and its adjacent regions. The red lines indicate fault locations sourced from Deng et al. (2003).

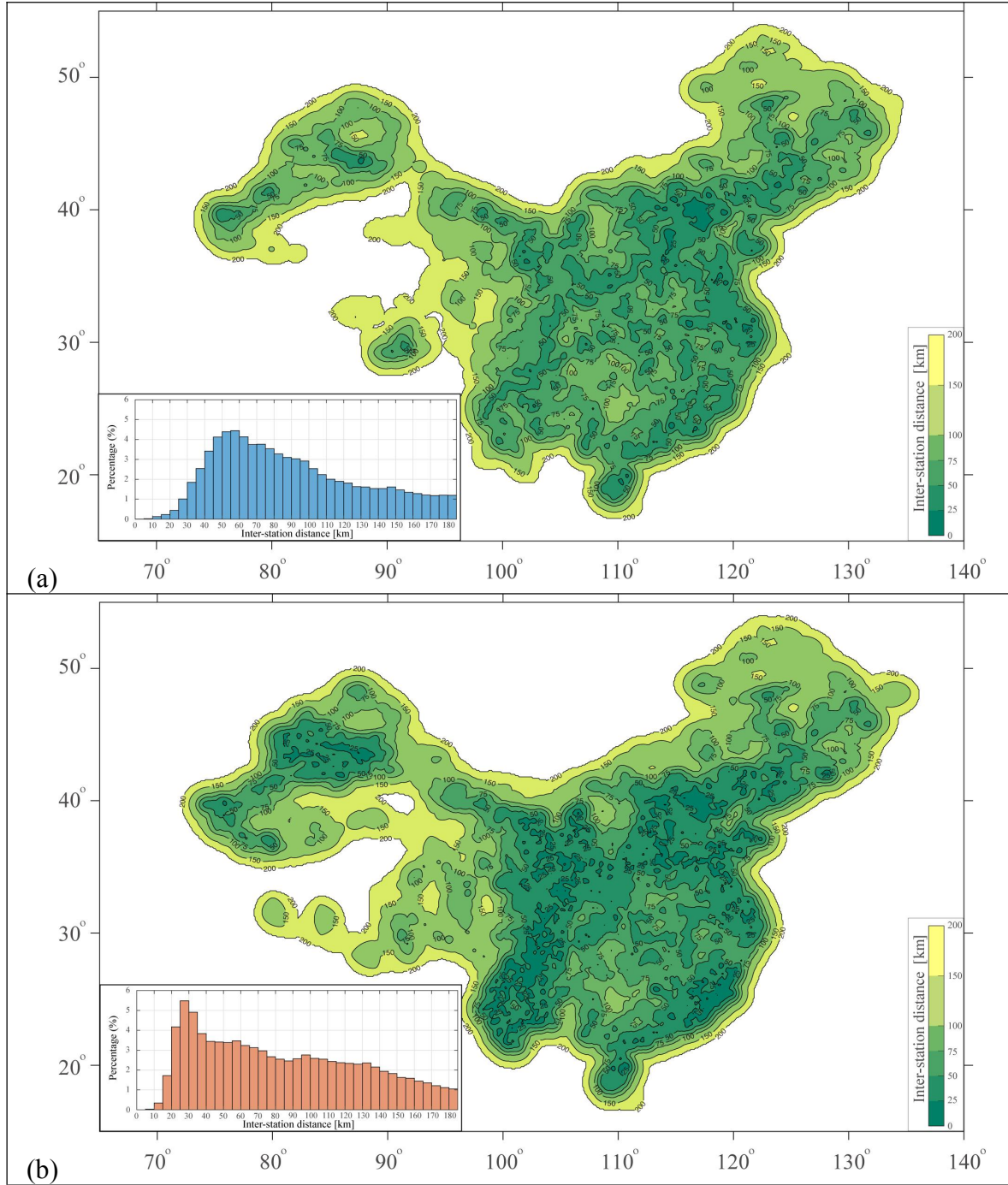


Figure S2. Maps showing the inter-station distances for the (a) existing and (b) planned networks. For any given site (assuming a grid of $0.1^\circ \times 0.1^\circ$ resolution), the inter-station distance is calculated as the average distance to the three closest stations (Kuyuk and Allen, 2013; Li and Wu, 2016; Li et al., 2021). The histograms in lower left show the binned distribution of the inter-station distances inside Chinese mainland.

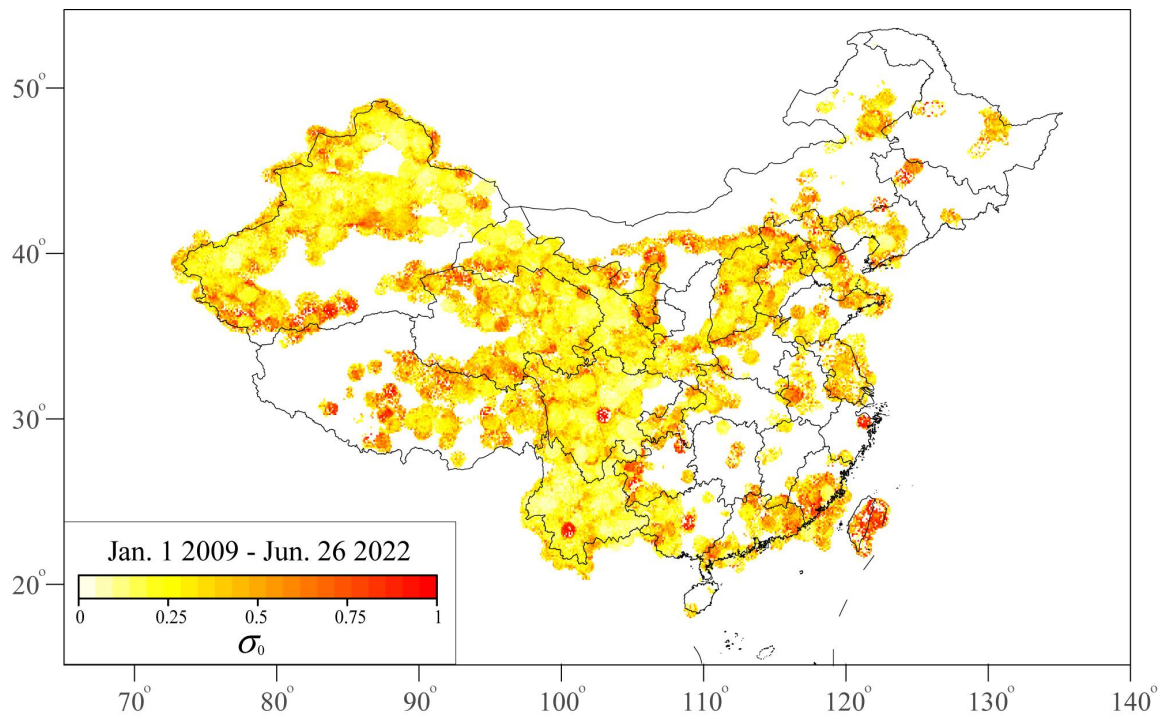


Figure S3. Map of the standard deviation of the observed M_c for the time period from 2009 to 2022. Same as Figure 2.

Reference

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